

An Attention-Guided Lightweight CNN–ViT Fusion Network for Potato Leaf Disease Classification Across Controlled and Natural Environments

Masuma Akter Akhi
Department of Software Engineering
Daffodil International University
Dhaka, Bangladesh

Md. Mizanur Rahman
Department of Software Engineering
Daffodil International University
Dhaka, Bangladesh

Fahmida Shah Nitu
Department of Software Engineering
Daffodil International University
Dhaka, Bangladesh

Abstract—Deep learning models for potato leaf disease can look almost perfect when training and testing are carried out on the same curated image source. This paper examines how much of that performance survives a shift to natural field imagery. Four compact models were trained only on the potato subset of PlantVillage: EfficientNet-B0, DeiT-Tiny, a direct CNN–ViT concatenation model, and an attention-guided CNN–ViT fusion model with image-specific branch weights. The selected checkpoints were then frozen and evaluated on 15,260 retained PLDD-UP images, with no target-domain tuning. To reduce dependence on a single random run, each model was trained with seeds 42, 123, and 2026. Internal Macro-F1 was effectively saturated ($0.9993\text{--}1.0000$), whereas external performance fell sharply. DeiT-Tiny achieved the strongest external Macro-F1 (0.6238 ± 0.0311), followed by direct concatenation (0.5903 ± 0.1264), attention fusion (0.5374 ± 0.0838), and EfficientNet-B0 (0.4866 ± 0.0506). The attention model improved over the CNN baseline but did not surpass the transformer-only or direct-fusion alternatives. Statistical testing, ablation, efficiency measurements, class-wise error analysis, and Grad-CAM point to the same practical issue: domain shift, rather than fusion complexity alone, remains the main limitation.

Keywords—potato leaf disease, CNN–ViT, vision transformer, attention-guided fusion, domain shift, external validation, Grad-CAM

I. INTRODUCTION

Image-based plant disease recognition is attractive because a leaf photograph can be collected quickly and analyzed without laboratory equipment. Public datasets and transfer learning have made this task easy to reproduce, and many papers now report accuracy close to 99% on curated benchmarks. The difficulty is that a clean benchmark and a farm photograph are not the same visual problem. PlantVillage established an important controlled benchmark [1], and Mohanty et al. later showed that a classifier reaching 99.35% on held-out PlantVillage images could fall to 31.4% on images from other sources [2]. That early result is still a useful warning: same-domain accuracy can exaggerate the level of practical robustness.

Potato disease classification exposes this problem clearly. Early Blight and Late Blight create visually meaningful symptoms, but field images add cluttered backgrounds, shadows, partial leaves, multiple leaves, variable camera quality, and changes in scale. Convolutional neural networks

(CNNs) are naturally strong at local texture and spatial patterns, whereas Vision Transformers (ViTs) can model longer-range relationships among image patches [3]–[5]. These different inductive biases have encouraged hybrid CNN–Transformer designs that attempt to capture both local and global information.

Recent work supports the value of hybridization but also shows that results depend strongly on the evaluation setting. ConvTransNet-S combined local perception and lightweight self-attention and achieved 98.85% on PlantVillage but 88.53% on a 10,441-image field dataset [10]. MobileViT-DAP used attention-enhanced MobileViT for six rice disease classes captured under field conditions [11]. ViTCon processed CNN and transformer representations in parallel and reported strong multiclass accuracy across corn, rice, and wheat [12]. For potato specifically, PLDNet combined DenseNet with a transformer attention module and reported 99.54% on PlantVillage and 87.50% on a separate Mendeley dataset [8]. LSeTNet further illustrates the current emphasis on compact hybrid architectures, external validation, and explainable AI [13].

This leads to a more practical question: if model development is restricted to a controlled source domain and the target field domain is left completely untouched, which representation transfers most reliably? The original research plan focused on five points: compact architecture design, the value of attention-guided fusion over ordinary concatenation, controlled-to-natural generalization, computational efficiency, and interpretability. The four-model experiment was built so these questions could be tested under one fixed data split and identical evaluation logic.

The paper makes four main contributions. First, PlantVillage is used only as the development domain, whereas PLDD-UP is preserved as an independent natural-field test set. Second, CNN-only, ViT-only, direct fusion, and sample-wise attention fusion are compared under the same protocol. Third, every architecture is repeated with three seeds and is evaluated with paired statistical tests rather than a single favorable run. Fourth, the comparison includes parameter count, approximate FLOPs, latency, error patterns, and Grad-CAM, so the discussion is not reduced to accuracy alone.

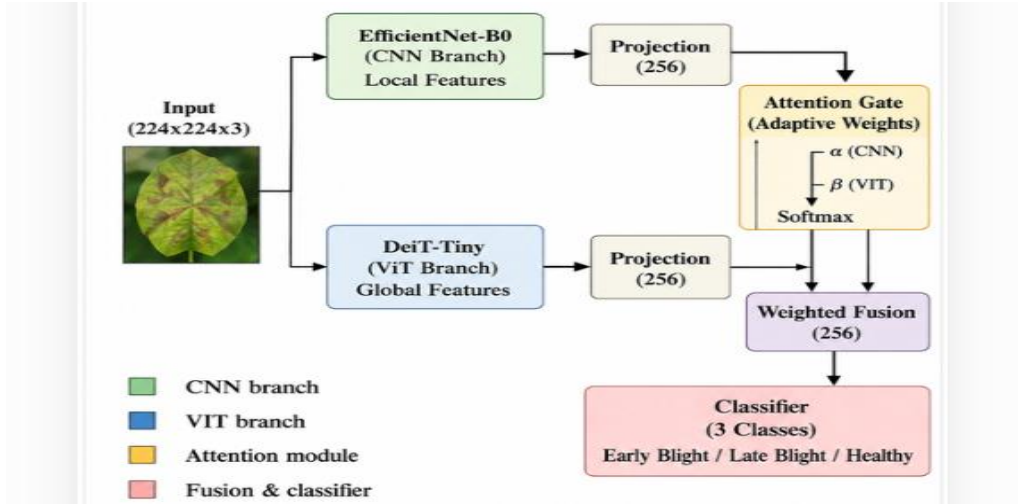


Fig. 1. Proposed sample-wise attention-guided CNN-ViT architecture. The two feature branches are projected to a common 256-D space, weighted by a softmax gate, and fused before three-class prediction.

II. RELATED WORK

A. Controlled benchmarks and external validity

Curated leaf datasets remain valuable because they support repeatable training and comparison, but they can hide acquisition-specific shortcuts. Ahmad et al. found that plant-disease recognition changes materially between controlled and field conditions and that data/background interventions can improve transfer [7]. The practical implication is simple: if the intended use case is a field photograph, a random split from the same curated source is not enough evidence of field readiness.

B. Local and global visual representations

EfficientNet introduced compound scaling to balance depth, width, and resolution in convolutional networks [3]. EfficientNet-B0 is relatively compact and provides a useful local-feature baseline. ViT replaces convolutional processing with patch embeddings and self-attention [4], while DeiT demonstrated that compact transformers can be trained and fine-tuned more efficiently than the original ViT recipe suggested [5]. Hybrid models use both representations because local lesion texture and global leaf context can carry complementary information.

C. Fusion, attention, efficiency, and explainability

Many hybrid networks merge branch features by concatenation or by sequential integration. Attention-based alternatives attempt to learn which information deserves more emphasis. This is appealing, but the extra flexibility can also

learn source-domain preferences. Recent field-oriented work therefore evaluates not only accuracy but also model size, computation, and robustness [10], [11], [13]. Explainability is another part of this discussion. Grad-CAM produces a class-discriminative localization map from gradients entering a convolutional layer [6]. In this study it is used as a debugging tool for the CNN branch rather than as proof that the full hybrid model is causally interpretable.

III. MATERIALS AND METHODS

A. Dataset roles and class harmonization

The two datasets were assigned different roles by design. The source domain was the potato subset of PlantVillage [1], containing relatively clean, centered leaf images. It was used for training, validation, and internal testing. The external target domain was PLDD-UP [9], collected from operational potato fields in Uttar Pradesh, India with digital cameras and smartphones under natural daylight. The published PLDD-UP release contains 15,519 images across Early Blight, Late Blight, and Healthy classes; after the project verification and cleaning pipeline, 15,260 images remained in the external-test manifest.

Class naming differed across the two sources, so folder labels were harmonized to Early_Blight, Late_Blight, and Healthy. Raw source folders were kept untouched. All later notebooks operated from verified manifests, which preserved the original files while recording cleaning and splitting decisions reproducibly.

TABLE I
FINAL DATASET SPLIT DISTRIBUTION

Split	Early	Healthy	Late	Total
External Test	4743	4448	6069	15260
Internal Test	150	23	150	323
Train	700	106	700	1506
Validation	150	23	150	323

B. Verification, cleaning, and split integrity

Before model development, both datasets were checked for expected class folders, readable images, dimensions, file counts, and label mappings. Hash-based duplicate analysis and near-duplicate review were used to identify repeated content and conflicting-label cases. Cleaning decisions were

written to metadata rather than applied destructively to the raw folders.

The fixed PlantVillage split contained 1,506 training images, 323 validation images, and 323 internal-test images. The external set contained all 15,260 retained PLDD-UP images. The same split was reused for every architecture and every seed. PLDD-UP was never used to select augmentation

settings, tune hyperparameters, choose an epoch, or select a checkpoint.

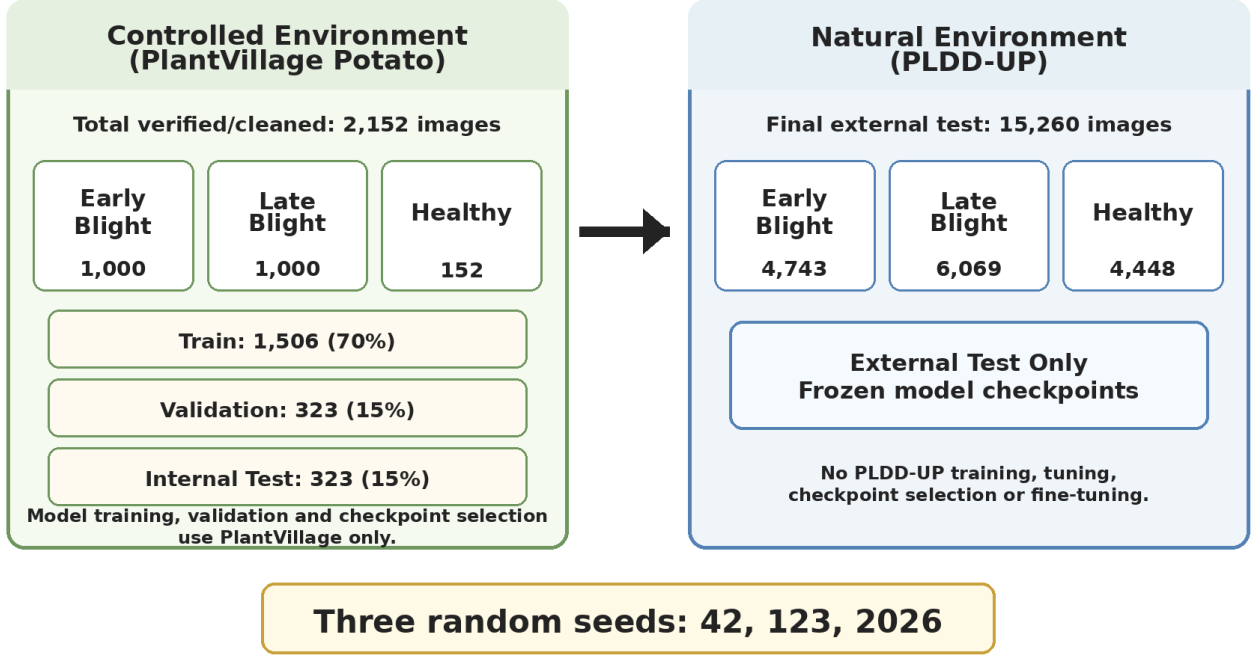


Fig. 2. Controlled-to-natural experimental workflow. PlantVillage is the only model-development domain; PLDD-UP remains an external test domain throughout.

C. Preprocessing and augmentation

Inputs were processed at 224×224 pixels. Training images were resized and then augmented with random resized cropping, horizontal flipping, rotation, moderate color jitter, and low-probability random erasing. Validation and test transforms were deterministic and used resizing, center cropping, tensor conversion, and ImageNet normalization. The Healthy class was much smaller in PlantVillage, so class-weighted cross-entropy was used to reduce the influence of frequency imbalance during optimization.

D. Baseline models

The CNN baseline was EfficientNet-B0 [3]. Its convolutional feature extractor was followed by adaptive global pooling, dropout, and a three-class linear classifier. The transformer baseline was DeiT-Tiny [5], fine-tuned for the same classes. Both single-branch models were initialized from ImageNet-pretrained weights.

A direct-fusion baseline was built with the same two backbones. The pooled EfficientNet-B0 vector and the DeiT-Tiny global representation were concatenated. The joint vector then passed through Layer Normalization, dropout, a 512-unit fully connected layer with GELU activation, another dropout layer, and the three-class classifier. This model provides the necessary comparison for asking whether the attention gate adds value beyond simply making both representations available.

E. Proposed attention-guided fusion

The proposed architecture changes the fusion rule while keeping the backbones matched to the concatenation baseline. CNN and ViT features are projected separately to 256 dimensions using a linear layer, Layer Normalization, GELU, and dropout. The two projected vectors are concatenated and processed by a small gating network that

outputs two logits. Softmax converts them into a CNN weight and a ViT weight for each input image.

$$z = \alpha_c c + \alpha_v v, \quad \alpha_c + \alpha_v = 1. \quad (1)$$

In (1), c and v denote the projected CNN and ViT features. The fused 256-dimensional representation z is passed through Layer Normalization, dropout, a 256-unit GELU hidden layer, another dropout layer, and the final classifier. The gate therefore adapts branch contribution at the sample level rather than applying a single fixed ratio to the entire dataset.

F. Training and repeatability protocol

All models used AdamW with cosine annealing, a maximum of 25 epochs, and early stopping with patience six. Learning rates were 1×10^{-4} for EfficientNet-B0, 5×10^{-5} for DeiT-Tiny, and 3×10^{-5} for both hybrid models. Weight decay was 1×10^{-4} . Single-branch runs used batch size 16; hybrid multi-seed runs used batch size 8. The best checkpoint was selected only by validation Macro-F1.

To reduce dependence on one initialization and one augmentation sequence, the complete experiment was repeated with seeds 42, 123, and 2026. The dataset split itself remained fixed. This makes seed-to-seed changes attributable mainly to training stochasticity rather than to different test samples. Final results are reported as mean $\pm$ sample standard deviation.

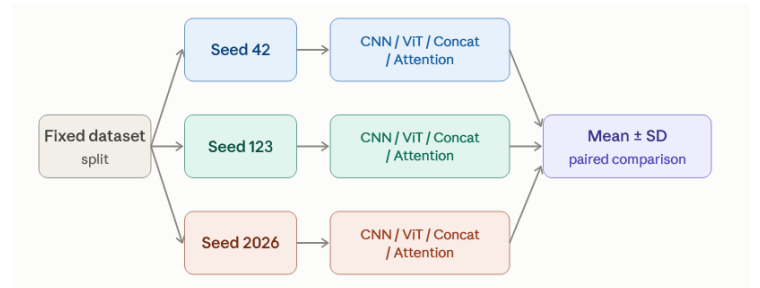


Fig. 3. Three-seed repeatability design used for all four architectures.

G. Evaluation, statistical analysis, and efficiency

Accuracy, macro-precision, macro-recall, and Macro-F1 were computed for the internal and external test sets. Macro-F1 was the primary ranking metric because each class contributes equally despite the source-domain imbalance. The generalization gap was defined as internal Macro-F1 minus external Macro-F1.

Saved predictions on the same external images enabled paired model comparison. Exact McNemar tests were performed for the proposed model against each baseline, seed by seed. In addition, 1,000 paired bootstrap resamples were used to estimate uncertainty for differences in accuracy and Macro-F1. Grad-CAM [6] was generated from the CNN branch of the proposed model on representative correct and incorrect PLDD-UP samples.

Efficiency was benchmarked using the seed-42 best checkpoint for each model on a Tesla T4 GPU with batch size

one and synthetic 224×224 RGB input. Twenty warm-up passes were followed by 100 timed forward passes. The reported latency therefore excludes image decoding, disk I/O, and preprocessing. Parameter count, estimated weight storage, approximate GFLOPs, throughput, and peak GPU memory were recorded separately.

IV. RESULTS

A. Controlled-domain performance

The internal PlantVillage test was effectively saturated. EfficientNet-B0, DeiT-Tiny, and direct concatenation achieved 1.0000 ± 0.0000 for accuracy, precision, recall, and Macro-F1. The attention-guided model was only marginally lower, with 0.9990 ± 0.0018 accuracy and 0.9993 ± 0.0013 Macro-F1. Within the controlled source domain, these results provide almost no practical separation among the four architectures.

TABLE II
THREE-SEED PLANTVILLAGE INTERNAL-TEST PERFORMANCE

Model	Accuracy	Precision	Recall	Macro F1
CNN + ViT + Attention	0.9990 ± 0.0018	0.9993 ± 0.0013	0.9993 ± 0.0013	0.9993 ± 0.0013
CNN only (EfficientNet-B0)	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
CNN + ViT Concatenation	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000
ViT only (DeiT-Tiny)	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000	1.0000 ± 0.0000

B. Natural-field external performance

External testing changed the ranking substantially. DeiT-Tiny achieved the best mean accuracy, 0.6352 ± 0.0215 , and the best Macro-F1, 0.6238 ± 0.0311 . Direct concatenation

ranked second in Macro-F1 at 0.5903 ± 0.1264 , although the large standard deviation shows clear seed sensitivity. The proposed attention model achieved 0.5685 ± 0.0578 accuracy and 0.5374 ± 0.0838 Macro-F1. EfficientNet-B0 was lowest at 0.4999 ± 0.0378 accuracy and 0.4866 ± 0.0506 Macro-F1.

TABLE III
THREE-SEED PLDD-UP EXTERNAL-TEST PERFORMANCE

Model	Accuracy	Precision	Recall	Macro F1
CNN + ViT + Attention	0.5685 ± 0.0578	0.6677 ± 0.0293	0.5430 ± 0.0634	0.5374 ± 0.0838
CNN only (EfficientNet-B0)	0.4999 ± 0.0378	0.5626 ± 0.0209	0.5275 ± 0.0309	0.4866 ± 0.0506
CNN + ViT Concatenation	0.6157 ± 0.0799	0.6884 ± 0.0378	0.5996 ± 0.0973	0.5903 ± 0.1264
ViT only (DeiT-Tiny)	0.6352 ± 0.0215	0.7006 ± 0.0032	0.6155 ± 0.0287	0.6238 ± 0.0311

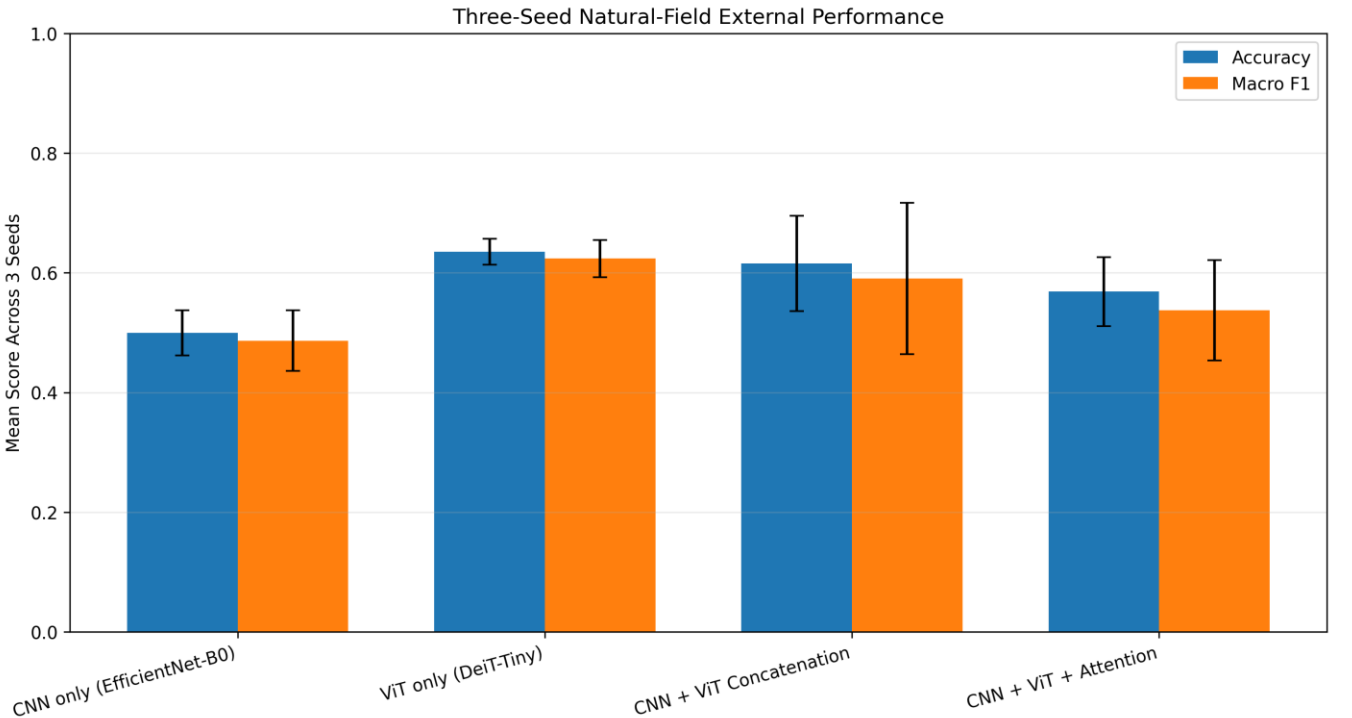


Fig. 4. Three-seed external accuracy and Macro-F1 on PLDD-UP. Error bars show the variation across seeds.

C. Controlled-to-natural generalization gap

The field drop is large for every architecture. EfficientNet-B0 lost 0.5134 Macro-F1 points between the internal and external tests. The attention model lost 0.4619, direct

concatenation lost 0.4097, and DeiT-Tiny had the smallest drop at 0.3762. The best field model was therefore also the model that degraded least when acquisition conditions changed.

TABLE IV
CONTROLLED-TO-NATURAL GENERALIZATION GAP

Model	Int. F1	Ext. F1	F1 drop	Acc. drop
CNN only (EfficientNet-B0)	1.0000	0.4866	0.5134	0.5001
ViT only (DeiT-Tiny)	1.0000	0.6238	0.3762	0.3648
CNN + ViT Concatenation	1.0000	0.5903	0.4097	0.3843
CNN + ViT + Attention	0.9993	0.5374	0.4619	0.4305

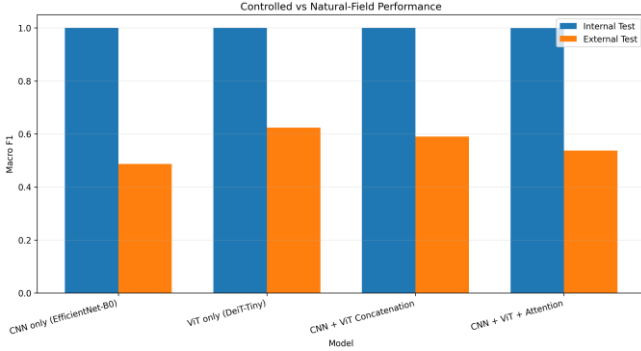


Fig. 5. Internal versus external Macro-F1. The controlled benchmark is nearly saturated, while every architecture declines on natural-field images.

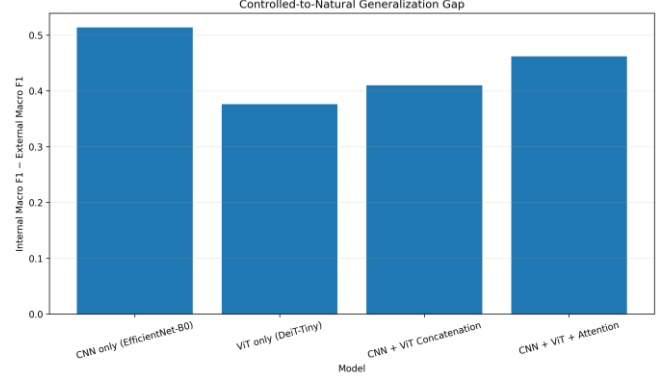


Fig. 6. Controlled-to-natural Macro-F1 drop for each architecture.

D. Paired statistical comparison

The paired tests sharpen the model ranking. Relative to EfficientNet-B0, the proposed attention model improved mean external accuracy by 0.0686 and Macro-F1 by 0.0508, and McNemar testing was significant for all three seeds.

Against direct concatenation, however, attention was lower by 0.0472 accuracy and 0.0529 Macro-F1 on average, with significant McNemar results in all three runs. Compared with DeiT-Tiny, attention was lower by 0.0667 accuracy and 0.0864 Macro-F1; two of the three seed-wise McNemar comparisons were significant, while one was not. The evidence therefore supports a gain over the CNN-only baseline, not a general superiority claim over transformer-based alternatives.

TABLE V
PROPOSED ATTENTION MODEL VERSUS EXTERNAL BASELINES

Baseline	Δ Acc.	Δ F1	Min p	Max p	Sig. seeds
CNN only (EfficientNet-B0)	+0.0686	+0.0508	1.5e-92	1.5e-14	3
CNN + ViT Concatenation	-0.0472	-0.0529	7.0e-311	8.8e-66	3
ViT only (DeiT-Tiny)	-0.0667	-0.0864	2.3e-219	0.59	2

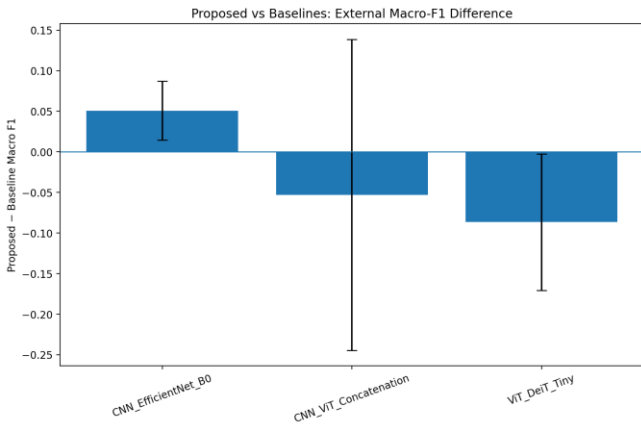


Fig. 7. External Macro-F1 difference of the proposed attention model relative to each baseline.

E. Architectural ablation

The four-model design also acts as an architectural ablation. A CNN-only path establishes the local-feature baseline. Replacing it with DeiT-Tiny raises external Macro-F1 by about 0.1372. Direct concatenation also improves over the CNN baseline, but with much greater seed-to-seed variation. Adding the attention gate reduces parameter count slightly compared with direct concatenation—10.11 M versus 10.29 M—yet field accuracy and Macro-F1 both decrease. Under this source-only training protocol, the gate appears to add optimization sensitivity without improving the cross-domain representation.

TABLE VI
ARCHITECTURAL ABLATION SUMMARY

Model	Fusion	Ext. Acc.	Ext. F1	F1 SD	F1 drop
CNN only (EfficientNet-B0)	None	0.4999	0.4866	0.0506	0.5134
ViT only (DeiT-Tiny)	None	0.6352	0.6238	0.0311	0.3762
CNN + ViT Concatenation	Concatenation	0.6157	0.5903	0.1264	0.4097
CNN + ViT + Attention	Attention	0.5685	0.5374	0.0838	0.4619

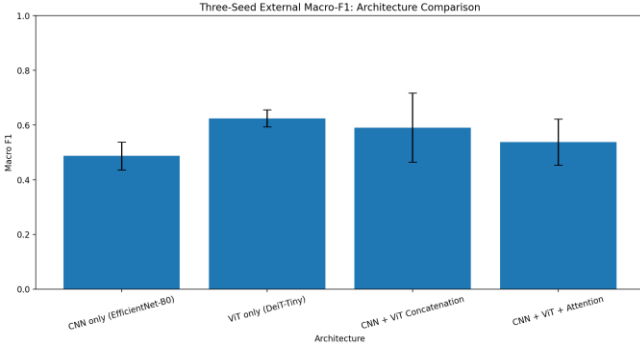


Fig. 8. External Macro-F1 across the four architectural configurations.

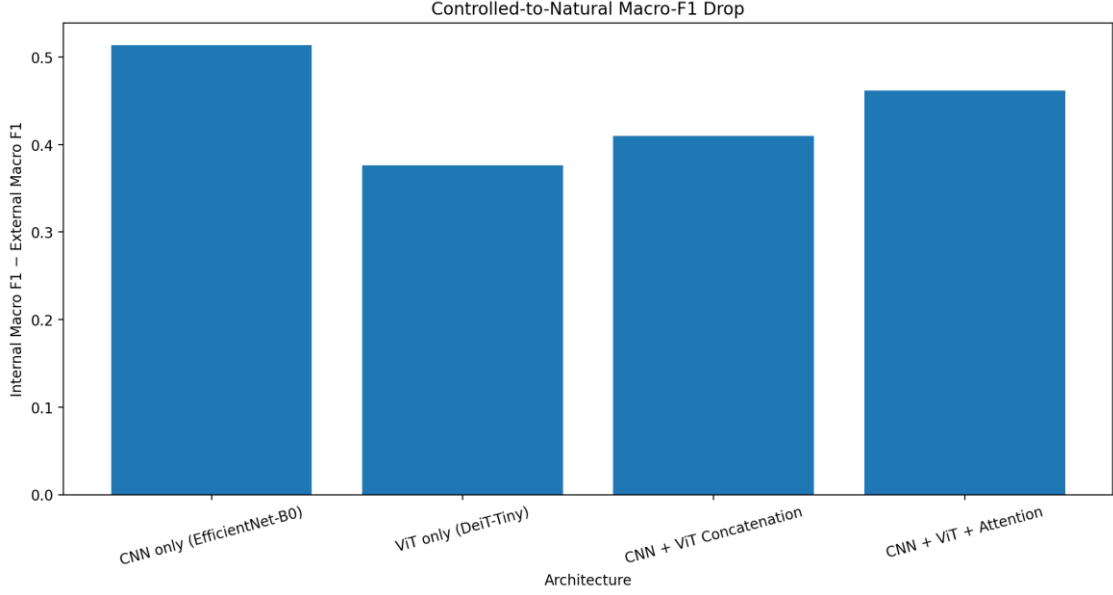


Fig. 9. Generalization drop viewed as an architectural ablation.

TABLE VII
SEED-42 EFFICIENCY BENCHMARK ON TESLA T4

Model	Params	MB	GFLOPs	Latency ms	img/s
CNN only (EfficientNet-B0)	4.01 M	15.46	0.40	19.40	51.6
ViT only (DeiT-Tiny)	5.52 M	21.08	1.08	5.51	181.4
CNN + ViT Concatenation	10.29 M	39.42	1.48	13.30	75.2
CNN + ViT + Attention	10.11 M	38.72	1.48	13.59	73.6

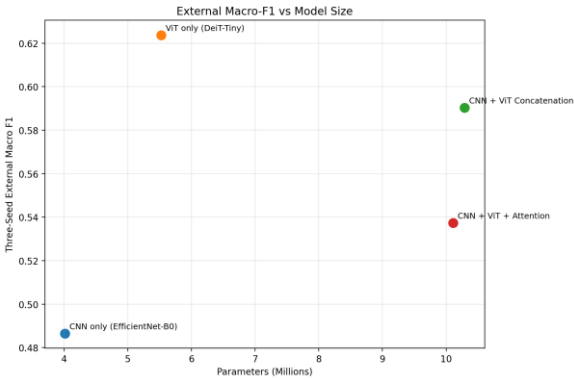


Fig. 10. External Macro-F1 versus parameter count.

G. Error analysis and explainability

The proposed model did not fail evenly across classes. Late Blight was the most reliable class, Healthy had the highest mean error rate, and Early Blight fell between them. This

F. Efficiency

Efficiency results add a practical dimension to the comparison. EfficientNet-B0 was the smallest model at 4.01 M parameters and approximately 0.40 GFLOPs. DeiT-Tiny used 5.52 M parameters and about 1.08 GFLOPs, but it was the fastest model in the measured T4 batch-one benchmark at 5.51 ms per forward pass and 181.4 images/s. The two hybrid models were close to 10 M parameters and 1.48 GFLOPs, with mean latency around 13–14 ms and weight storage close to 39 MB. These timings are hardware-specific, but they show that the best field model was also compact in this experimental setting.

pattern is plausible given the source split: only 106 Healthy images were available for training, compared with 700 Early Blight and 700 Late Blight images. Weighted loss can change the penalty assigned to minority-class mistakes, but it cannot manufacture the missing visual variety of healthy leaves under field conditions.

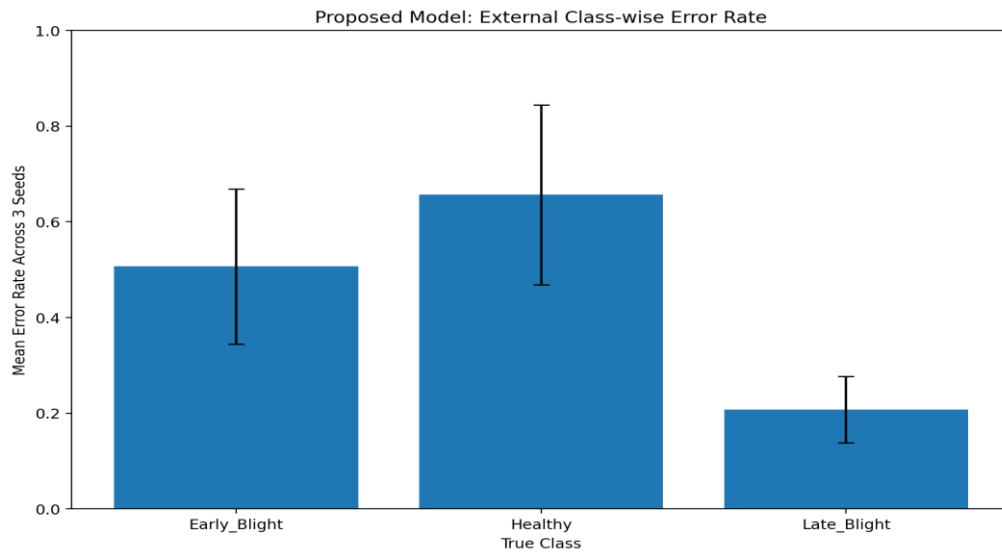


Fig. 11. Class-wise external error rate of the proposed attention model across three seeds.

Representative Proposed-Model External Misclassifications

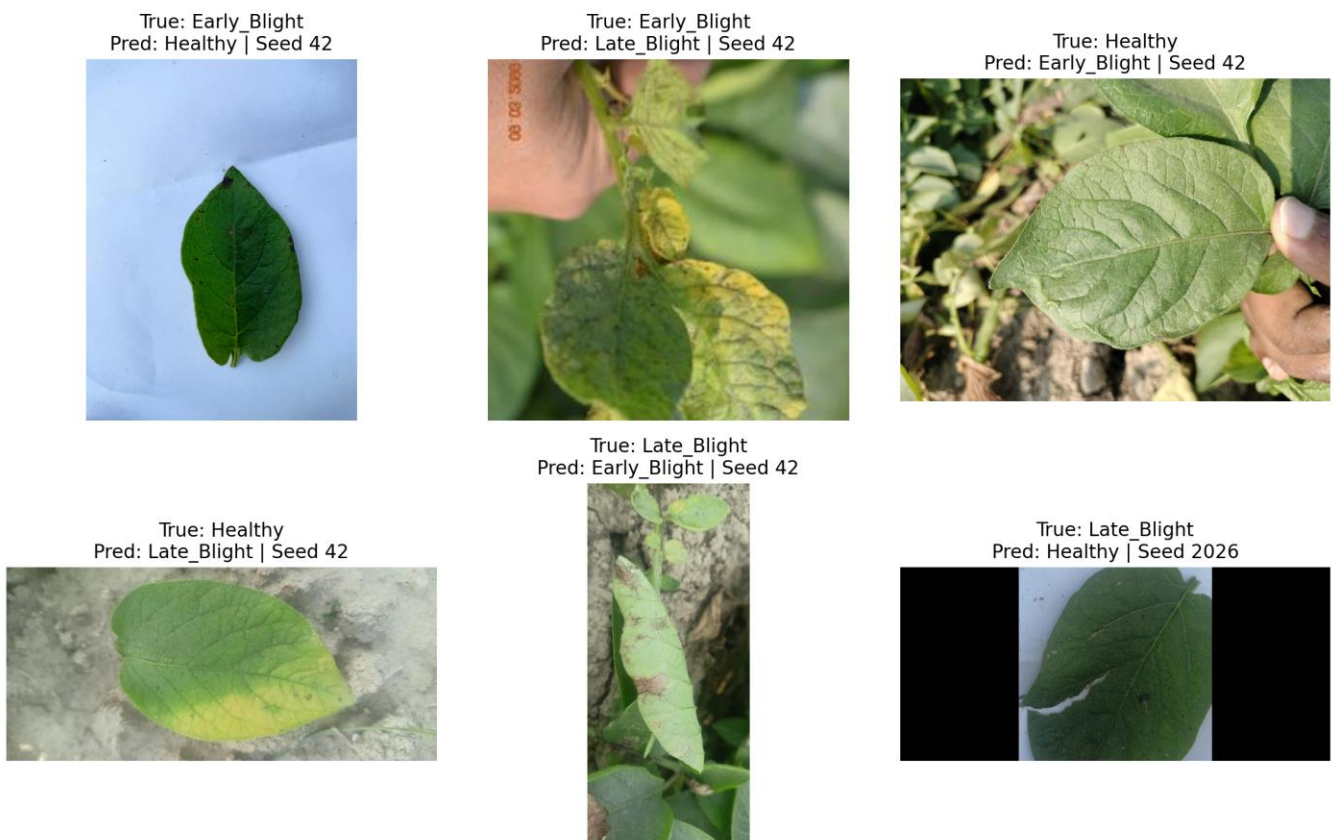


Fig. 12. Representative high-confidence misclassifications from PLDD-UP. The selected examples illustrate class confusion under natural backgrounds and varied leaf presentation.

Grad-CAM maps were generated from the CNN branch for both correct and incorrect predictions and inspected together with the learned branch weights. Several cases showed concentrated activation over visibly affected leaf regions; other cases were diffuse or included background areas. The

visualization is intentionally interpreted as branch-level diagnostic evidence. Because Grad-CAM was not applied to the transformer tokens, it is not a complete explanation of the full two-branch decision.

Representative Grad-CAM examples from the attention model (CNN branch)

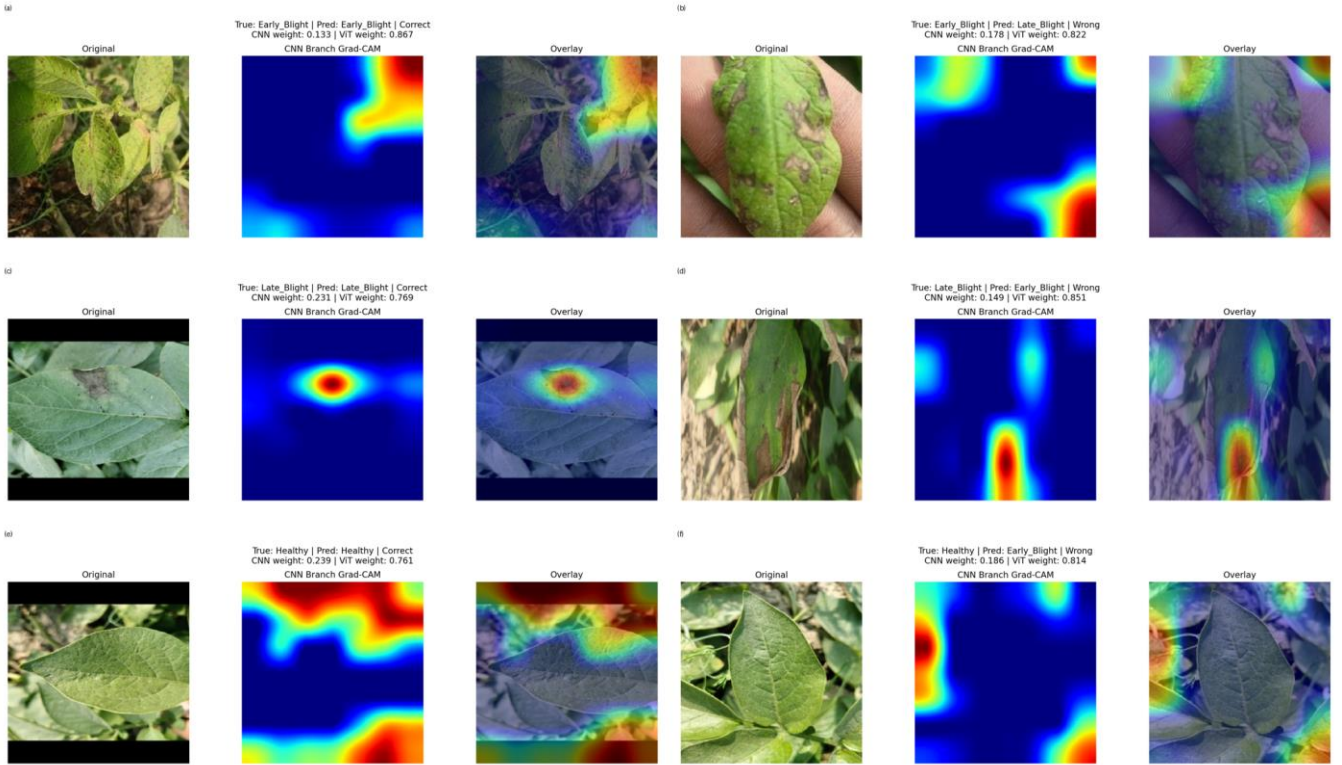


Fig. 13. Six representative Grad-CAM cases from the CNN branch of the proposed attention model. Each case contains the original image, CNN activation map, and overlay.

V. DISCUSSION

The most important result is the size of the acquisition-domain gap. On PlantVillage, every architecture looked essentially solved. Once the same frozen checkpoints were moved to PLDD-UP, the ranking separated and Macro-F1 spread from 0.4866 to 0.6238. The controlled benchmark is therefore useful for repeatable model development, but it is not sufficient for selecting the most field-robust architecture.

Among the four models, DeiT-Tiny stands out because it was both the strongest and the most stable on the external field set. A reasonable interpretation is that global patch interactions were less tied to the regular source background than the local convolutional representation. Field-oriented hybrid studies such as ConvTransNet-S similarly motivate combining local and global modeling in cluttered environments [10]. Still, the present experiment does not independently manipulate background, lesion size, or texture. The explanation should therefore be treated as a hypothesis rather than a causal conclusion.

The attention gate did not behave as initially expected. It was designed to choose a better local-global balance for each image. The transformer branch clearly helped, because attention fusion outperformed EfficientNet-B0. Yet the gate did not surpass DeiT-Tiny or direct concatenation. One plausible reason is that the gate itself was learned only from PlantVillage. If source images contain highly regular visual cues, a flexible weighting mechanism can become another place for source-specific specialization. Adaptivity within one domain is not the same thing as invariance across domains.

Direct concatenation reveals a different part of the story. Its mean external Macro-F1 was second highest, but its standard deviation of 0.1264 was far larger than DeiT-Tiny's

0.0311. A single run could therefore give a misleading impression of superiority. This is why the three-seed design matters: it exposes not only the average level of performance but also the sensitivity of the model to stochastic optimization.

The efficiency benchmark also changes how the deployment result should be read. The proposed hybrid remains modest compared with large ViT families, but it roughly doubles the parameters of the two single-branch baselines. In the measured environment, the additional complexity did not buy the best field score. For a practical screening application, the present evidence would favor DeiT-Tiny as the zero-adaptation candidate. A production system would still require calibration and additional field-focused development before it could be treated as reliable agricultural decision support.

For that reason, the next step should address the data shift directly instead of adding another fusion block. Field-aware augmentation, broader collection of healthy leaves, self-supervised pretraining on unlabeled field images, domain-adversarial training, and carefully separated domain-adaptation data are all plausible directions. PLDD-UP should not be repeatedly tuned against without creating a new independent final test set, because doing so would convert the benchmark into a validation domain.

VI. LIMITATIONS

Several limitations constrain the conclusions. The PlantVillage source set is small and strongly imbalanced, particularly for Healthy leaves. PLDD-UP provides a much more realistic target environment, but it represents one geographic region and one growing season and cannot cover every cultivar, camera, disease severity, or lighting condition. The study also measures zero-adaptation transfer by design;

that is a demanding generalization test but not necessarily the best achievable deployment strategy.

Three seeds are more informative than one run, yet they remain a modest sample of optimization variability. Grad-CAM was generated only from the CNN branch, so transformer attention rollout or token-attribution analysis would provide a more complete explanation of the hybrid. Finally, the latency benchmark uses synthetic batch-one inputs on one Tesla T4 and excludes preprocessing and I/O. It should be read as a controlled comparison under a stated setup, not as a universal hardware ranking.

VII. ANSWERS TO THE RESEARCH QUESTIONS

RQ1 asked whether a compact CNN-ViT design could achieve high potato disease performance while controlling complexity. The answer is domain-dependent: all compact models were extremely strong internally, but the hybrid did not retain the best field performance. RQ2 asked whether attention-guided fusion improves on conventional concatenation. Under this experiment it did not; direct concatenation had higher mean external Macro-F1. RQ3 concerned controlled-to-natural generalization, and the results show a substantial drop for every model, with DeiT-Tiny showing the smallest decline.

RQ4 asked whether the proposed architecture offers competitive efficiency. The attention model remained relatively compact at 10.11 M parameters and 1.48 GFLOPs, but the single-branch DeiT-Tiny achieved both better external performance and lower measured latency. RQ5 concerned interpretability. Grad-CAM provides useful qualitative inspection of the CNN branch and exposes several plausible lesion-focused and background-associated responses, but the study does not provide a quantitative transformer-side explanation. The interpretability claim should therefore remain limited to diagnostic visualization rather than a comparative proof of explanation quality.

VIII. CONCLUSION AND FUTURE WORK

This study evaluated four compact potato leaf disease classifiers across a controlled source domain and an independent natural-field target domain. EfficientNet-B0, DeiT-Tiny, direct CNN-ViT concatenation, and attention-guided CNN-ViT fusion all reached approximately perfect performance on PlantVillage. Their external behavior was very different. DeiT-Tiny achieved the strongest and most stable PLDD-UP Macro-F1 at 0.6238 ± 0.0311 . Direct concatenation ranked second, the attention-guided model ranked third, and EfficientNet-B0 ranked fourth.

The proposed attention mechanism therefore produced a useful but negative architectural result. It significantly improved on the CNN baseline, which confirms that the transformer branch carried valuable information, but the learned gate did not outperform simpler transformer-based alternatives. This changes the emphasis of the work in a productive way: the main challenge is not obtaining another 99% score on a controlled benchmark, but maintaining disease discrimination when background, lighting, pose, and acquisition conditions change.

Future work should focus on field-aware representation learning, better Healthy-class diversity, target-independent regularization, domain adaptation with a separate adaptation

split, calibration, broader multi-location evaluation, and transformer-specific explainability. A deployment-oriented extension could compare server-based inference with compressed or ONNX-converted models on mobile hardware, but any deployment claim should be supported by a new independent field test.

DATA AVAILABILITY AND REPRODUCIBILITY

PlantVillage is publicly available through the PlantVillage project [1]. PLDD-UP is publicly available from Mendeley Data, Version 1, DOI 10.17632/3j4nfvkp2n.1 [9]. The experiment records dataset verification, cleaning decisions, fixed split manifests, random seeds, best checkpoints, prediction CSV files, statistical comparisons, Grad-CAM outputs, efficiency metadata, and final result tables. These artifacts should be archived with the final submission where licensing permits.

REFERENCES

- [1] D. P. Hughes and M. Salathé, "An open access repository of images on plant health to enable the development of mobile disease diagnostics," [arXiv:1511.08060](https://arxiv.org/abs/1511.08060), 2015.
- [2] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, art. 1419, 2016, doi: [10.3389/fpls.2016.01419](https://doi.org/10.3389/fpls.2016.01419).
- [3] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. 36th Int. Conf. Machine Learning (ICML)*, PMLR, vol. 97, pp. 6105–6114, 2019.
- [4] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2021.
- [5] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. 38th Int. Conf. Machine Learning (ICML)*, PMLR, vol. 139, pp. 10347–10357, 2021.
- [6] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Computer Vision (ICCV)*, pp. 618–626, 2017, doi: [10.1109/ICCV.2017.74](https://doi.org/10.1109/ICCV.2017.74).
- [7] A. Ahmad, A. El Gamal, and D. Saraswat, "Toward generalization of deep learning-based plant disease identification under controlled and field conditions," *IEEE Access*, vol. 11, pp. 9042–9057, 2023, doi: [10.1109/ACCESS.2023.3240100](https://doi.org/10.1109/ACCESS.2023.3240100).
- [8] A. Mondal, A. Chatterjee, and N. Avazov, "A hybrid CNN-transformer model with adaptive activation function for potato leaf disease classification," *Scientific Reports*, vol. 16, art. 4282, 2026, doi: [10.1038/s41598-025-34406-4](https://doi.org/10.1038/s41598-025-34406-4).
- [9] P. K. Singh, A. Yadav, D. Yadav, S. Tiwari, and A. Chandel, "PLDD-UP: Potato Leaf Disease Dataset from Uttar Pradesh, India," *Mendeley Data*, ver. 1, 2026, doi: [10.17632/3j4nfvkp2n.1](https://doi.org/10.17632/3j4nfvkp2n.1).
- [10] S. Jia, G. Wang, H. Li, Y. Liu, L. Shi, and S. Yang, "ConvTransNet-S: A CNN-Transformer hybrid disease recognition model for complex field environments," *Plants*, vol. 14, no. 15, art. 2252, 2025, doi: [10.3390/plants14152252](https://doi.org/10.3390/plants14152252).
- [11] M. Zhang, Z. Lin, S. Tang, C. Lin, L. Zhang, W. Dong, and N. Zhong, "Dual-attention-enhanced MobileViT network: A lightweight model for rice disease identification in field-captured images," *Agriculture*, vol. 15, no. 6, art. 571, 2025, doi: [10.3390/agriculture15060571](https://doi.org/10.3390/agriculture15060571).
- [12] P. K. Verma, N. Gupta, A. K. Sharma, N. Rakesh, and M. Gulhane, "ViTCon: A hybrid CNN-ViT model for improved plant leaf disease detection," *Cogent Food & Agriculture*, vol. 11, no. 1, art. 2562165, 2025, doi: [10.1080/23311932.2025.2562165](https://doi.org/10.1080/23311932.2025.2562165).
- [13] J. Ahmmed, M. A. Kabir, A. ur Rehman, and A. Bermak, "A lightweight hybrid CNN and transformer model for medicinal leaf disease classification with explainable AI," *Scientific Reports*, vol. 16, art. 8243, 2026, doi: [10.1038/s41598-026-39182-3](https://doi.org/10.1038/s41598-026-39182-3).